

EDUCATION

Apr. 2021 – present **B.Eng. in Computer Science and Engineering**
Expected graduation: Mar. 2026

Waseda University, Tokyo, Japan

SELECTED FULL PAPERS

1. **Rikiya Takehi**, Fernando Diaz, Tetsuya Sakai. 2025
Diversification as Risk Minimization
International Conference on Web Search and Data Mining (**WSDM 2026**) (acceptance rate: 16%).
Best Paper Award (1/799, top 0.1%)
[preprint](#) | [preprint](#)
2. Ren Kishimoto*, **Rikiya Takehi***, Koichi Tanaka, Masahiro Nomura, Riku Togashi, Yuta Saito. 2025.
Beyond Match Maximization and Fairness: Retention-Objectified Two-Sided Matching
International Conference on Learning Representations (**ICLR 2026**) (acceptance rate = 28%)
[preprint](#) | [Openreview](#)
3. **Rikiya Takehi**, Kosuke Kawakami, Masahiro Asami, Yuta Saito. 2025
General Framework for Off-Policy Learning with Partially-Observed Reward.
International Conference on Learning Representations (**ICLR 2025**) (acceptance rate: 32%).
[arxiv](#) | [OpenReview](#) | [presentation](#) | [poster](#)
4. **Rikiya Takehi**, Ellen M. Voorhees, Tetsuya Sakai, Ian Soboroff. 2025
LLM-Assisted Relevance Assessments: When Should We Ask LLMs for Help?
SIGIR Conference on Research and Development in Information Retrieval (**SIGIR 2025**) (acceptance rate: 21%).
[arxiv](#) | [slides](#) | [code](#)
5. **Rikiya Takehi**, Akihisa Watanabe, and Tetsuya Sakai. 2023.
Open-Domain Dialogue Quality Evaluation: Deriving Nugget-level Scores from Turn-level Scores
SIGIR in the Asia Pacific Region (**SIGIR AP 2023**) (acceptance rate: 36%).
[proceedings](#) | [arxiv](#) | [code](#) | [poster](#) | [slides](#)
6. **Rikiya Takehi**, Benjamin Clavié, Sean Lee, Aamir Shakir. 2025.
Fantastic (small) Retrievers and How to Train Them: mxbai-edge-colbert-v0 Tech Report.
[Tech Report](#) | [Blog](#) | [17M ColBERT model](#) | [32M ColBERT model](#) → **1.5M+ downloads within 3 months**
7. **Rikiya Takehi***, Koichi Tanaka*, Ren Kishimoto, Masahiro Nomura, Riku Togashi, Yuta Saito. 2025.
Objective-driven Calibrated Recommendations
Submitted to ICML 2026.
[preprint](#)
8. Benjamin Clavié, Sean Lee, **Rikiya Takehi**, Aamir Shakir, Makoto P. Kato. 2025.
Simple Projection Variants Improve ColBERT Performance
[arxiv](#)

SELECTED SHORT PAPERS

1. Tetsuya Sakai, Khant Myoe Rain, **Rikiya Takehi**, Sijie Tao, Youngin Song.
Open-Source LLM-based Relevance Assessment vs. Highly Reliable Manual Relevance Assessment: A Case Study
Conference on Information and Knowledge Management (**CIKM 2025**) (short paper; acceptance rate: 30%).
[proceedings](#)

HONORS & AWARDS

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|-----------|--|
| 2026–2028 | Toyota Riken PhD Fellowship (link)
Awarded to outstanding early-career researchers, providing full funding for two years of graduate school , including tuition, a monthly living stipend, travel expenses, health insurance, and application-related costs, plus up to three years of partial funding support. |
| 3/2025 | Project Research Award, Waseda University
Recognized for exemplary research activity (awarded to 2 out of 300+ students in the department). |

EXPERIENCE

- 3/2026 - **Research Intern, NVIDIA Trustworthy AI Team** Remote
- My boss: [Yusuke Hirota](#) & [Frank Wang](#).
 - Will work on **Agentic AI**. PhD-level position.
 - First undergraduate/master's student from a Japanese university ever to do a research internship at NVIDIA.
- 2/2025 - 9/2025 **Research Collaboration, CMU Language Technology Institute** Pennsylvania, USA - Remote
- (Bi-)weekly meetings with [Prof. Fernando Diaz](#).
 - Proposed and led research on minimization of missing information in a retrieved / recommended ranking (full paper at **WSDM 2026, Best Paper Award**).
- 10/2023 - 8/2024 **Guest Researcher, NIST - National Institute of Standards and Technology** Maryland, USA
- Conducted research in the retrieval group with **Dr. Ian Soboroff & Dr. Ellen Voorhees**.
 - Led work on *evaluation of generated text via nugget-based evaluation*.
 - Proposed and led research on the efficient usage of LLMs along with limited manual annotations for building test collections. (full paper at **SIGIR 2025**)
- 8/2025 - 2/2026 **Research Intern, Mixedbread AI** California, USA - Remote
- Led development of new approach to improve the performance and efficiency of late interaction models.
 - Built two open-source ColBERT models. SOTA of its model size.
- 9/2024 - 2/2026 **Research Intern, CyberAgent AI Lab.** Tokyo, Japan
- Proposed and led research on a framework to optimize the categorical proportions, instead of relying on a heuristic like calibrated recommendations (under review)
 - Proposed and led research an algorithm on online dating applications, where we maximize user retention, using real-world dataset of the company (under review)
- 9/2023 - 3/2025 **Research Intern, Hakuhodo Tech. Inc.** Tokyo, Japan
- Proposed and led research on a framework that leverages secondary rewards to help the optimization for **off-policy learning for partially observed rewards**. (full paper at **ICLR 2025**.)
- 7/2023 - 09/2025 **Research Intern, Hanjukukaso Inc.** Tokyo, Japan - remote
- Advisor: **Yuta Saito**
 - Working on **fairness in ranking** and **off-policy evaluation & learning**.
- 4/2023 - Present **Student Researcher, Sakai Lab - Waseda Univ.** Tokyo, Japan
- Advisor: **Prof. Tetsuya Sakai**
 - Proposed and led research mainly on **evaluation of IR**. (full paper at **SIGIR 2025**)
 - Proposed and led research on **evaluation of dialogue systems**. (paper at **SIGIR AP 2023**)

OPEN-SOURCE PROJECTS

- **mxbai-edge-colbert-v0 series**
 - SOTA small ColBERT model for its model size. (**17M outperforms ALL <1B model retrievers on LongEmbed benchmark**.)
 - [17M model](#) and [32M model](#) achieved **1M+ downloads in two months**.

LEADERSHIP & PROFESSIONAL SERVICE

- **Co-organizer of the TREC Product Search and Recommendations Track 2025** ([website here](#)) with **Drs. ChengXiang Zhai** (UIUC), **Michael Ekstrand** (Drexel), **Surya Kallumadi** (Lowe's), and **Daniel Campos** (Snowflake)
- **Invited Panelist at NTCIR 2025** with **Profs. Maarten de Rijke** (UvA), **Mark Sanderson** (RMIT), **Charles Clarke** (UWaterloo), and **Ian Soboroff** (NIST).

INVITED TALKS

- 07/2025 **SIGIR eCom Workshop 2025**
- Hosted by **Dr. Tracy Holloway King** (Adobe)
 - *Product Search and Recommendations* (with **Prof. ChengXiang Zhai**)
- 06/2025 **EVIA 2025**
- Hosted by **Prof. Charles Clarke** (UWaterloo)
 - *Using LLMs as Assistants for Building Large Test Collections*

- 11/2024 **National Institute of Informatics**
- Hosted by **Prof. Noriko Kando** (NII)
 - *Using LLMs as Assistants for Building Test Collections (+Trends and Problems of Test Collections)*
- 08/2024 **University of Maryland College Park**
- Hosted by **Prof. Douglas Oard** (UMD)
 - *Nugget-Based Evaluation and the Use of LLMs*

SKILLS

- Programming: Python (Expert), C++, Java, JavaScript, HTML/CSS
- Frameworks: PyTorch, TensorFlow, Hugging Face Transformers, NumPy, pandas
- Japanese - native, English - fluent (**TOEFL iBT 114**), French - proficient (DELF/CEFR B2)